

Find the Boundary

What you're doing

You're testing an image-grounding AI model. You upload (or supply by URL) any image, type a short prompt about something in that image, run the model, and then **judge whether the model's answer is correct**. When it's wrong, you fix it.

Every submission is one of:

- **A "break"** — you found a case where the model is wrong, and you corrected it.
- **A "pass"** — you found a case where the model is right.

The goal isn't to "win" or to "lose" — it's to **find the capability boundary** of the model: where does it succeed, where does it fail, and what does the cliff edge look like? A confident, honest judgment on either side is what we want.

The three output formats

You pick the format at run time. Each one asks the model a different kind of question.

Format	Model's Job	Example Prompt
Bounding box	Draw a tight box around the object(s) the prompt describes.	the red mug to the left of the laptop
Point	Place a single point on the object the prompt describes.	the recycle button on the screen
Counting	Count the objects matching the prompt and return one point per instance.	the orange fish swimming behind the rocks

Pick the format that matches what you actually want the model to do. Don't switch formats mid-image to try to game a "fail" — that's noise, not signal.

Writing prompts

Good prompts are **specific, falsifiable, and grounded in what's actually visible**. The model can't read your mind, but it should reliably handle anything an attentive human can confirm from the image.

Do

- **Use specific references:** the small red button next to the search bar, not the button.
- **Stack constraints when stress-testing:** the second cat from the right that is **NOT** looking at the camera.
- **Use spatial relations:** left of, above, closest to, partially hidden by.
- **Use negation deliberately:** the dog that is **NOT** wearing a collar.
- **Reference attributes:** color, size, orientation, material, state (open, cracked, running).

Don't

- **Don't reference things that aren't in the image.** If the answer is "there's no such thing", that's a malformed prompt — discard the round.
- **Don't write open-ended prompts.** **Describe the picture** isn't a grounding prompt; **the largest object in the picture** is.
- **Don't write ambiguous prompts** where two or more objects equally satisfy the constraints. If the answer depends on your mood, the prompt is broken.
- **Don't use NSFW, PII, or copyrighted content** in your images. If you wouldn't post it on a work Slack channel, don't upload it.
- **Don't probe the model with images of identifiable people** unless they're public figures in a clearly public context.

Aim for hard prompts AND hard images — that's where the value is

A model that gets every easy prompt on every easy image right doesn't tell us anything new. The most useful submissions come from **probing the edge of capability** along two axes at once: the image you pick, and the prompt you write on top of it. A well-chosen image with a sharp prompt is far more valuable than an easy image with a tricky prompt — and *much* more valuable than the reverse.

A good heuristic: if you'd predict the model has a **50–70% chance** of getting the (image, prompt) pair right, that's a great submission. 99% is easier than we need. 1% is probably out-of-distribution rather than boundary-probing — interesting once, but a stream of those is also low signal.

Hard images to look for

- **Dense / cluttered scenes** — crowds, supermarket shelves, parking lots, food on a table, toolboxes, dense text, fish tanks, bird flocks. Many similar-looking instances in close proximity.
- **Small targets in context** — UI screenshots with tiny icons, a button on a busy toolbar, fine print on a label, a small bird against foliage, a pedestrian in a wide street scene. (Use the magnifier loupe.)
- **Low contrast / camouflage** — white object on a white background, a moth on bark, a stop sign at night, a transparent bottle on a glass shelf.
- **Heavy occlusion** — the target is half-hidden by another object, behind glass with reflections, peeking out from behind a person, partially cut off by the frame.
- **Unusual viewpoint or framing** — top-down (drone, X-ray, microscope), extreme oblique angles, fisheye, mirror reflections, rotated 90°/180°, partial crops.
- **Visual interference** — motion blur, glare, harsh shadows, water reflections, neon-lit scenes, very high or low key lighting, photographs of a screen / monitor.
- **Repetitive patterns and identical objects** — racks of identical bottles, egg cartons, a grid of windows on a building, identical chairs in a cafeteria — where ordinal references ("the third from the left") matter.
- **Text-heavy scenes** — receipts, document scans, road signs, infographics, game UI. The model often confuses text content with object identity.

- **Out-of-domain images** — medical (X-rays, MRIs), industrial (PCB closeups, factory floors), satellite, microscopy, art and stylized images, hand-drawn diagrams, screenshots of video games.
- **Multi-instance ambiguity** — multiple objects equally well-matching a literal prompt, forcing the constraint to disambiguate.

Hard prompt patterns

- **Multi-hop spatial reasoning** — "the cup to the left of the lamp that is closest to the window", "the second-tallest building behind the bridge".
- **Negation and exclusion** — "the dog that is NOT wearing a collar", "every bottle except the green one", "the only chair without anyone sitting on it".
- **Ordinal references** — "the third book from the right", "the fifth person in the second row from the back".
- **Counting under stress** — many similar instances, some partially occluded or clustered, where the model has to actually distinguish individuals.
- **Subtle attribute distinctions** — "the very dark red shirt" vs. "the dark brown shirt" in the same image, "the matte cup" vs. "the glossy cup", "the chair that is rotated".
- **Constraint stacking** — three constraints in one prompt: shape AND color AND position. Models often satisfy two and drop one.
- **Affordance / state references** — "the door that is ajar", "the cup that's about to spill", "the button that is pressed". Requires understanding state, not just appearance.
- **Cross-reference between objects** — "the person looking at the dog", "the book the cat is sitting on", "the cable that's plugged into the monitor on the right".

Pairing the two axes

The strongest submissions combine a hard image with a sharp prompt. For example:

- **A dense bottle shelf** + *"the third bottle from the left in the second row that is NOT green"* — small, ordinal, negated, in clutter.
- **A screenshot of a busy UI** + *"the close button on the dialog box that is in the background, not the modal in front"* — small target, occlusion, multi-instance ambiguity.

- A **flock of birds** + *"the bird in flight whose wings are pointed downward"* — counting-by-elimination + subtle state.

Don't be afraid to **fail and learn**: submit a prompt, see how the model breaks, then submit a tweaked version on the same image (use the Tweak This / boundary-pair flow) to find the exact constraint that flipped it. A pass/fail pair on the same image is one of the most useful kinds of submission you can make.

Judging the model — verdict rules

You have two verdicts.

I win — the model got it wrong

Pick this when **at least one** of these is true:

- **Wrong object.** The model boxed/pointed at something that doesn't match the prompt.
- **Missed object(s).** In counting mode, the model under-counted; in bbox mode, it missed instances.
- **Hallucinated object(s).** The model returned more items than the image contains, or invented something that isn't there.
- **Loose box.** The box doesn't tightly fit the object — significant padding, or cuts off the object. (For bbox mode, "tight" means within ~5% of the visible silhouette on each side.)
- **Wrong point.** The point lands clearly off the object — outside it, on the wrong instance, or on a non-salient part.
- **Format violation.** The model returned malformed JSON, the wrong key, an out-of-bounds coordinate, or the wrong number of items for the prompt.

When you pick **I win**, you correct the answer:

1. **Fix the boxes / points on the canvas** — drag to move, drag handles to resize, click to delete, click in empty space to add. Use the **magnifier loupe** (top-right button or **M** key) for small objects.
2. **Optionally edit the answer text** (click the pencil) — useful if the answer is structurally malformed or the counting label is wrong.
3. **Optionally fix the thinking trace** (click the pencil on the trace box) — if the model's reasoning was wrong in a specific, citable way.

4. **Add tags** describing why it failed (Failure Type) and how you challenged it (Challenge Strategy).
5. **Optionally write a one-line failure description.**
6. **Rate your confidence** on both the trace and the answer.

AI wins — the model got it right

Pick this when the answer is correct **as-is**:

- The boxes are tight ($\leq 5\%$ slack on each edge), or the points are on the correct object.
- The count is right (in counting mode).
- No missed and no hallucinated items.
- The format is valid.

You can still:

- **Edit the trace** — even if the answer is right, the reasoning might be wrong (e.g. the model cited the wrong landmarks but stumbled into the right answer). A trace fix on a correct answer is one of the more valuable things you can contribute — it shows the model how to *get to* the right answer for the right reasons.
- **Skip canvas edits** — by design, on a pass the answer is locked. If the boxes are loose enough to bother you, that's actually a **fail** — not a "pass with corrections".

When you pick **AI wins**, **rate your trace confidence**. (Answer confidence is N/A on a pass — the verdict itself is your answer judgment.)

When in doubt

If you're genuinely 50/50, treat it as a **fail** and correct what you'd want the model to have done. A confident correction is more useful than a coin-flip pass.

Editing the canvas

The image canvas is your scratchpad for fixing the model's output. The same tools work whether you're in bbox, point, or counting mode (point and counting share the point tool).

Action	How
Add a box	Click + drag on empty canvas while the answer pencil is on.
Add a point	Single-click on empty canvas.
Move a box / point	Drag it.
Resize a box	Drag a corner handle (visible when the box is selected).
Delete	Select, press Backspace / Delete.
Cycle through overlapping items	Repeated clicks at the same spot cycle through overlapping boxes / points
Magnifier loupe	Click the  5× toggle in the canvas top-right, or press M. Follows the cursor; shows pixels around the cursor with a crosshair + live draft rect. Toggle off to free the screen
Revert all canvas edits	Click the "Revert" button next to the answer box.
Undo / redo	Standard Ctrl/Cmd-Z / Ctrl/Cmd-Shift-Z.

For tiny objects (favicons, single icons, small badges) the magnifier is the recommended workflow — turn it on, hover over the target, drag the box at 5× zoom for pixel-accurate edges.

Tagging — when and how much

Tags describe *what* happened so similar cases can be grouped together later. Pick the tags that genuinely describe what you saw — leaving a category empty is fine if nothing fits.

- **Failure Type** — when you picked **I win**, what kind of failure was it?
(**SmallObjectMiss**, **RelationSwap**, **NegationBlindness**, etc.)
- **Challenge Strategy** — what was your attack?
(**AmbiguousReference**, **SmallTargets**, **MultiHopRelation**, **DenseOverlap**, etc.)
- **Image Type** — what kind of image is this?
(**IndoorScene**, **ClutteredScene**, **TextOnObjects**, etc.)

Use the **core** list by default. Only reach for the **full** list when no core tag fits — picking overly-specific tags makes the same case look different from other similar cases.

Boundary pairing (Tweak This)

On the Challenge page you can pick a previous submission on the same image and **probe its boundary** — submit a new prompt that's a deliberate variation. The intent is to find the edge: if **the red mug** works, does **the red mug to the left of the laptop** still work? If **the second cat** fails, does **the second cat from the right** succeed?

When you do this:

- The new round is linked to the previous one so the two can be looked at together later.
- If you predicted you'd flip the verdict and the verdict didn't flip, the UI will surface a soft "boundary didn't move" warning — feel free to submit anyway; both outcomes are useful.

Don't force pairings just for the badge. A useful pair is one where the second prompt is a *meaningful* tweak of the first.

Confidence (1–5 Likert)

After every submission you'll be asked one or two confidence sliders. They mean:

- **Trace confidence** — how sure are you that the chain-of-thought is correct? (If you didn't edit the trace: how sure are you the model's reasoning was right? If you did: how sure are you about your fix?)
- **Answer confidence** (*only on fail*) — how sure are you about the corrected answer? (If you didn't make canvas edits: how sure are you that your judgment "the answer is wrong" is correct?)

Value	Meaning
5	I'm certain.
4	I'm pretty sure.
3	I lean toward this but could see the other side.
2	I'm honestly guessing more than knowing.
1	I have no idea — flip a coin.

Be honest. A confident *wrong* answer is more harmful than an honest low-confidence one — there's no penalty for clicking 1 or 2 when you genuinely aren't sure.

Common mistakes

- **"Looks roughly right" passes.** A box with 15% padding around the object is a **fail**, not a **pass**. The model has a 5% tightness bar.
- **Counting-by-default in counting mode.** If the model returns 12 points and the image has 12 instances, *check each point* — sometimes the count is right but the points are on the wrong instances.
- **Switching formats to "win".** If you ran bbox and the model failed, don't switch to point and try the same prompt for an easier pass — that's noise. Resubmit in bbox or change the prompt.
- **Vague prompts then disagreement.** "the cat" with three visible cats is ambiguous. Either disambiguate the prompt or accept any valid answer.
- **Editing the answer on a pass.** If you find yourself wanting to fix the boxes on a pass, that's a fail. Flip the verdict.
- **Forgetting the magnifier.** Small UI elements (icons, badges, single letters) are exactly what the magnifier is for. Use it.

Engagement & badges

You'll see a session stats bar at the top of the page and badge unlock toasts. These are purely informational — streaks, daily activity, unique images, exploration depth, etc. They don't gate anything; the intent is just to make a long session feel less abstract.

Don't game the badges. A streak of hollow submissions is worse than a shorter streak of careful ones.

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